

Restaurant Success & Opportunity Map

Predictive Modeling and Market Opportunity Analysis for Philadelphia Restaurants

Executive Summary

Philadelphia has one of the most competitive restaurant market in the East Coast. With thousands of options across dozens of neighbourhoods, a natural question comes up: What actually makes a restaurant succeed here? Is it the food quality, the location, the price point or something else entirely? This project analyzes Yelp data for restaurants in Philadelphia to understand what drives restaurant success and to identify where new opportunities may exist in the market. The goal is two-fold: first, to build a predictive understanding of what factors influence whether a restaurant performs well, and second, to translate those insights into practical, business-oriented recommendations for decision-making.

To achieve this, we combined structured restaurant data (such as ratings, location, and attributes) with customer-generated data (reviews and check-ins). We began with exploratory analysis to understand general patterns in the dataset, followed by feature engineering to create meaningful variables such as customer engagement metrics and competition indicators. These features were then used to train machine learning models to classify whether a restaurant is likely to succeed.

A baseline Logistic Regression model was first implemented to establish a benchmark. To capture complex relationships between variables, we then applied a Random Forest model, which significantly improved performance across all evaluation metrics. Beyond predictive modelling, we also explored customer reviews using NLP techniques to better understand what drives customer satisfaction. This adds depth to the analysis by linking numerical performance with real customer feedback.

The results show that restaurant success is influenced by a combination of factors rather than a single variable. Customer engagement (measured through reviews and check-ins), geographic location, and the level of local competition all play important roles. High-performing restaurants tend to be located in areas with strong demand while maintaining a balance in competition density.

The project highlights neighborhoods with high demand but relatively lower competition, suggesting potential locations for new restaurants. It also emphasises the importance of early customer engagement and consistent service quality.

Overall, this project demonstrates how data-driven approaches can support strategic decisions in the restaurant industry. By combining machine learning with business understanding, we can move beyond intuition and make more informed, evidence-based choices.

1. Problem Definition & Framework

Business Problem

Opening a successful restaurant involves significant risk, and many establishments fail due to poor location decisions or a limited understanding of customer demands. This project aims to address the following business questions for a competitive market like Philadelphia.

- What factors influence restaurant success?
- Can we predict whether a restaurant is likely to succeed?
- Where are the potential opportunities for new restaurants?

Definition of Success

For our analysis, restaurant success is defined using two indicators.

- Whether the restaurant is currently operational
- The overall rating of the restaurant (based on star reviews)

As per the above definitions, the problem is framed as a classification task, where the goal is to predict whether a restaurant is likely to be successful.

Analytical Framework

The project follows a structured, three-step analytical approach:

- **Descriptive Analysis**
To explore the dataset and identify patterns in restaurant performance, including trends in ratings, customer engagement, and location.
- **Predictive Modeling**
To build machine learning models that can predict restaurant success based on relevant features.
- **Opportunity Detection**
To identify areas with high demand but relatively low competition, indicating potential opportunities for new restaurants.

Data Description

The analysis is based on Yelp data specifically filtered for Philadelphia restaurants. The dataset includes:

- Restaurant Data: Ratings, location, and business attributes
- Review Data: Customer reviews and associated ratings
- Check-in Data: Customer activity, used as a proxy for popularity

2. Data Preparation & Feature Engineering

Data Preparation

The dataset used in this analysis consists of restaurant-level information, customer reviews, and check-in data from Yelp. These datasets were first combined into a single modelling dataframe using `business_id` as the primary key.

To ensure consistency across datasets, key preprocessing steps were performed. Postal codes were standardised before merging, and review data were cleaned by handling missing text values. External census data was incorporated wherever available to capture neighbourhood-level characteristics such as population and median income. These variables provide additional context about the environment in which each restaurant operates.

After merging all datasets, missing values in review-based features were filled with zeros. This was done because missing values in this context typically indicate that a restaurant has no reviews rather than missing information. The final dataset represents each restaurant as a single row with all relevant features consolidated.

In addition to structured features, basic Natural Language Processing (NLP) techniques were explored to better understand customer feedback. Review text was preprocessed and transformed using TF-IDF to identify commonly occurring terms across different rating categories. While these text-based features were not directly included in the final predictive model to maintain simplicity, they provided useful qualitative insights into customer preferences and supported patterns observed in the structured data.

Feature Engineering

Several features were created to better capture restaurant performance and customer behaviour:

- **Review-based features:**
The number of reviews (`num_reviews`), average review rating, and average review length were computed to measure customer engagement and satisfaction. Additional metrics such as total “useful,” “funny,” and “cool” votes were included as indicators of review quality and interaction.
- **Demographic features:**
Census-based variables such as population and median income were included to account for location-specific

factors that may influence restaurant success.

- **Categorical encoding:**
Attributes such as WiFi availability, alcohol service, and noise level were converted into numerical form. These were encoded in a way that preserves their relative order (eg, free WiFi > paid > none), allowing them to be used effectively in machine learning models.

Feature Analysis

To understand how engineered features relate to restaurant success, we analyzed the distribution of review counts across open and closed restaurants.



Figure 1: Review Count vs. Restaurant Success

The boxplot shows that restaurants that are currently open tend to have higher median review counts compared to those that are closed. Open restaurants also exhibit a wider spread, including several high-value outliers, indicating strong customer engagement. In contrast, closed restaurants are concentrated at lower review counts.

This suggests that review volume is strongly associated with restaurant success, as it reflects both popularity and sustained customer interest. Based on this insight, the review count was retained as an important predictor in the modelling stage.

3. Modelling Approach & Results

Modeling Approach

To predict restaurant success, we framed the problem as a binary classification task, where the goal is to classify whether a restaurant is successful (`is_open = 1`) or not (`is_open = 0`). Based on the features engineered in the previous section, we applied two models: Logistic Regression as a baseline and Random Forest as an advanced model.

Before training the models, the dataset was split into training and testing sets using an 80–20 split to ensure that model performance is evaluated on unseen data. All selected features, including review-based metrics, encoded categorical variables, and demographic variables, were used as inputs.

Baseline Model: Logistic Regression

Logistic Regression was used as the baseline model due to its simplicity and interpretability. It assumes a linear relationship between the input features and the log-odds of the target variable.

The model performed reasonably well, providing a good starting point for comparison. However, it cannot capture complex relationships between features. For example, the effect of location combined with customer engagement may not be fully represented in a linear model.

Advanced Model: Random Forest

To address the limitations of the baseline model, we implemented a Random Forest classifier. Random Forest is an ensemble method that builds multiple decision trees and combines their outputs to improve prediction accuracy.

The model was configured with 200 trees and a maximum depth of 10 to balance performance and overfitting. One key advantage of Random Forest is its ability to capture nonlinear relationships and interactions between features, which are common in real-world datasets like this one.

Model Comparison

The performance of both models was evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score.

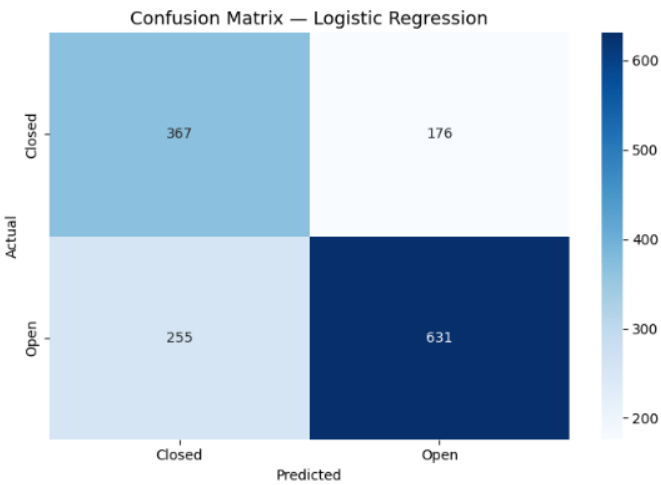
Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	~68%	~75%	~71%	~0.73
Random Forest	72.8%	78.3%	77.5%	0.779

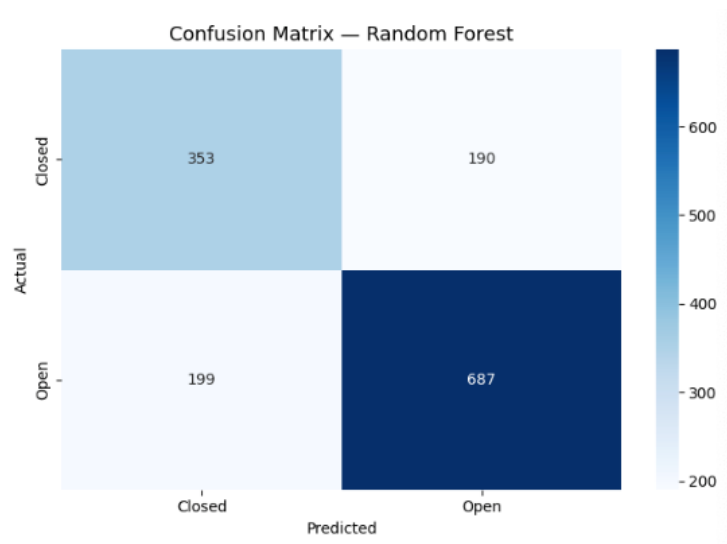
The Random Forest model outperformed Logistic Regression across all metrics. The most significant improvement was in recall, indicating that the model is better at identifying successful restaurants.

Diagnostic Analysis

To further evaluate model performance, we analysed prediction behaviour using diagnostic techniques. The Random Forest model showed better separation between classes compared to Logistic Regression, indicating stronger predictive capability.

A confusion matrix was used to visualise the model's predictions and errors. The results suggest that the model has a balanced performance, with relatively fewer false negatives compared to the baseline.



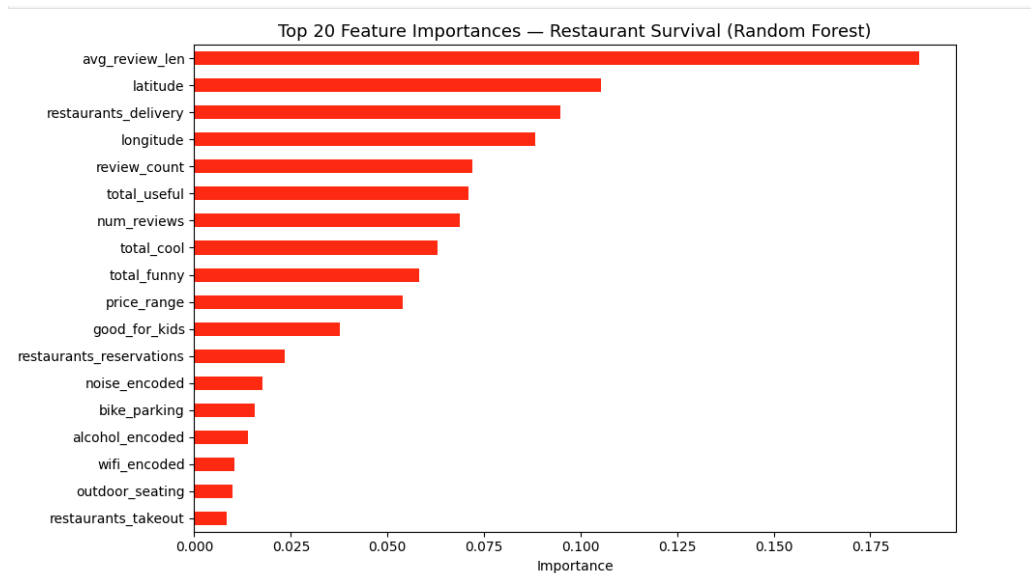


Model Interpretability

Understanding which features drive predictions is important for translating model results into business insights. Random Forest provides feature importance scores that indicate the contribution of each variable.

The most important features identified were:

- Number of reviews
- Check-ins (customer activity)
- Location-related features



This aligns with earlier observations from the feature analysis section, reinforcing the importance of customer engagement and geographic factors.

Why Random Forest Performed Better

The improved performance of Random Forest can be attributed to its ability to capture complex relationships between variables. For example, a restaurant in a high-income area with moderate competition may perform differently compared to a similar restaurant in a highly saturated area.

While Logistic Regression treats features independently, Random Forest can model these interactions, leading to better predictions.

Summary of Results

Overall, the modelling results show that restaurant success is influenced by multiple interacting factors rather than a single variable. The Random Forest model provides a more accurate and reliable prediction framework, making it suitable for identifying high-potential restaurant opportunities.

4. Key Findings & Business Recommendations

Key Findings

Based on the modelling results and feature analysis, several important insights emerge regarding what drives restaurant success.

1. Customer Engagement is the Strongest Predictor

The number of reviews and overall customer interaction metrics were consistently identified as the most important features in the model. Restaurants with higher review counts are significantly more likely to remain open. This suggests that sustained customer engagement is a strong indicator of long-term success.

2. Location Plays a Critical Role

Geographic and demographic factors, such as neighbourhood characteristics and local demand, have a noticeable impact on restaurant performance. Restaurants located in areas with higher population density and income levels tend to perform better.

3. Competition Has a Non-Linear Effect

While some level of competition indicates demand, excessive competition can negatively affect success. Restaurants in moderately competitive areas tend to perform better than those in highly saturated locations.

4. Customer Experience Drives Outcomes

Insights from review-based features suggest that customer satisfaction, reflected through ratings and engagement metrics, directly influences success. Restaurants with consistent positive feedback tend to maintain higher performance over time.

Recommendations

Based on these findings, the following recommendations are proposed:

1. Select Locations with High Demand and Moderate Competition

New restaurants should target areas where customer demand is strong but competition is not overly saturated. This balance increases the likelihood of attracting and retaining customers.

2. Focus on Early Customer Engagement

Restaurants should prioritise strategies that encourage customer interaction, such as promotions, loyalty programs, and a strong online presence. Increasing review volume early can significantly improve visibility and long-term success.

3. Monitor and Act on Customer Feedback

Regular analysis of customer reviews can help identify strengths and areas for improvement. Addressing negative feedback promptly can enhance customer satisfaction and retention.

4. Differentiate Through Experience, Not Just Offerings

Given the competitive nature of the market, restaurants should focus on service quality, ambience, and overall experience rather than relying solely on cuisine type.



Figure 2: Restaurant Demand vs. Customer Satisfaction

This visualization shows the relationship between customer demand (measured by review count) and customer satisfaction (average rating). Restaurants with both high demand and high ratings represent strong performers in the market. These establishments are likely to have sustained customer interest and positive experiences. On the other hand, restaurants with low demand and lower ratings may struggle to remain competitive. This analysis helps identify areas where new restaurants can enter the market by targeting high-demand segments while maintaining strong service quality.

5. Limitations & Future Directions

Limitations

While the analysis provides useful insights into restaurant success, there are several limitations that should be considered.

First, the definition of success in this project is based on whether a restaurant is currently open and its rating. While these are reasonable proxies, they do not fully capture business performance. Factors such as revenue, profitability, and customer retention are not included in the dataset, which limits the ability to measure true success.

Second, the data is sourced from Yelp, which may introduce bias. Yelp users represent a specific segment of customers, and their reviews may not reflect the opinions of the broader population. Additionally, restaurants with more online presence are more likely to receive reviews, which may skew the analysis toward more visible businesses.

Another limitation is that the analysis is largely static. The models do not explicitly account for changes over time, such as trends in customer preferences or seasonal effects. Restaurant performance can vary significantly over time, and this is not fully captured in the current approach.

Finally, some potentially important variables are missing or approximated. For example, competition was not directly measured using precise geographic proximity, and customer demand was inferred using review counts rather than actual foot traffic data.

Future Directions

There are several ways this project can be extended to improve both accuracy and practical usefulness.

One important direction is incorporating time-series analysis. By analyzing how reviews, ratings, and check-ins change over time, we can better understand growth patterns and predict future performance more accurately.

Another improvement would be to apply more advanced NLP techniques to the review text. Instead of using basic aggregated features, methods such as sentiment analysis or topic modeling could provide deeper insights into customer preferences and pain points.

Additionally, incorporating more detailed location-based features would strengthen the analysis. For example, measuring competition based on the number of nearby restaurants within a certain radius could provide a more accurate representation of market saturation.

Finally, including external business metrics such as revenue data, foot traffic, or online ordering trends would allow for a more comprehensive definition of success. This would make the model more aligned with real-world business outcomes.